A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 30

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - Steam washing (hot water / steam jets kill organisms in the sediment)	1
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Chemicals cause the oil molecules to join together / solidify **1** Producing more solid material that can be collected / lifted more easily than liquid oil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Air contains about 210 cm^3 of oxygen per litre whereas water at 20 °C holds a maximum of about 6 cm^3 per litre (less than 3% of the amount in air) **1** So a small amount of oxygen-consuming activity (decomposition, respiration) or warming can remove most of the oxygen – many aquatic organisms are near the lower end of their range of tolerance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **deoxygenation** → loss of dissolved oxygen from the water • **range of tolerance** → the range of conditions in which a species can survive	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 254-262; AQA spec 3.4.3.2.6-3.4.3.2.7.	
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** Metal-rich lumps 5-10 cm across on the deep seabed (4000-5000 m), containing about 30% manganese plus nickel, copper, cobalt, iron, titanium **1** Exploitation would be very expensive (remotely operated collectors, pumping to the surface) / no viable technology yet **1** International agreement on ownership of the seabed and its resources is needed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
02.2	**1 mark each** 1 mark for each of the 3 points below. **1** Collectors disturb the seabed and kill benthic organisms living on it **1** Separating nodules from sediment creates plumes that increase turbidity with unknown consequences **1** Re-deposition of sediment covers and kills filter feeders and other benthic organisms – slow recovery in the deep sea **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
02.3	**1 mark each** 1 mark for each of the 4 points below. **1** Used in small amounts in many essential technologies (smartphones, hybrid vehicles, wind turbines, fibre optics, LEDs) **1** Exploitable land deposits are limited and mainly in China – chemical separation is complex and produces large amounts of toxic waste **1 Bacterial adsorption:** dissolved rare earth ions adsorb onto the cell surfaces of bacteria (Bacillus subtilis) and are washed off at specific pHs to separate them **1** Deep sea sediments contain relatively high concentrations but no viable extraction method yet **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorption** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • 'wind turbines can generate all the time' - output depends on wind speed, and they shut down in very high winds	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 145-146; AQA spec 3.2.3.7.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** Trees absorb CO2 by photosynthesis and store carbon in wood as they grow **1** Once mature there is no further net storage (photosynthesis balanced by respiration / decomposition), though the standing forest remains a reservoir **1** Harvesting the timber and keeping it as wood (eg house construction) then replanting produces additional storage as the new trees grow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 4 points below. **1 Pre-combustion:** the fuel is modified before use so no CO2 is released when burnt - eg gasification of coal to hydrogen and CO2 (CO2 stored, hydrogen distributed) – oxy-fuel combustion in pure oxygen giving only CO2 and water vapour which is condensed out **1 Advantage:** capture at a large central facility – suits small dispersed sources (vehicles) that cannot capture CO2 **1 Post-combustion:** CO2 removed from exhaust gases by solvents, membrane filtration, adsorption / desorption, cryogenic separation, graphene **1 Disadvantage:** only about 20% of flue gas is CO2 so separation is costly / energy-intensive – none fully developed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **carbon capture** → capturing carbon dioxide from flue gases and storing it underground • **adsorption** → nutrient ions sticking to the surfaces of clay and humus	4
03.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Injected into depleted oil / gas fields or saline aquifers **1** Injected into oil reservoirs to provide pressure for secondary oil recovery **1** Reaction with fractured basalt to form solid carbonate minerals **1 Concern:** leakage back to the atmosphere / monitoring over centuries – limited suitable geology – cost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 151-152; AQA spec 3.2.4.2.	
04.1	**2 marks** 2 marks for the point below; award fewer where it is only partly made. **2** When there is a difference in level between the sea and the basin: on the ebb (about 6-8 hours) water flows out through the turbines – no generation when levels are equal (slack water) **OR** possible on the flood when the sea is higher than the basin **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Two periods of slack water in each 12.5-hour cycle (twice in about 25 hours) give no flow – the timing does not match demand **1** Tides depend on the positions of the Moon and Sun which are known, so output can be forecast years ahead and alternative supplies planned **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1 Spring tides:** Sun and Moon aligned so their gravitational effects combine giving a large tidal range – neap: effects oppose giving a small range **1** Energy depends on the mass of water and height difference, so output is much greater at springs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** When demand is low, surplus electricity pumps water into the lagoon above sea level **1** The stored water is released through the turbines later when demand is high (pumped storage) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **pumped storage** → water pumped uphill when demand is low and released to generate at peak	2
04.5	**1 mark each** 1 mark for each of the 2 points below. **1** Large tidal range (eg Severn Estuary) **1** Estuaries, bays or headlands / channels that focus the flow and increase velocity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 214-215; AQA spec 3.3.4.1.	
05.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B	1

Do not write outside the box

Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 3 points below. **1** TBT is highly toxic to marine molluscs / crustaceans and an endocrine disruptor altering growth and reproduction of oysters / whelks even at very low concentrations **1** It bioaccumulated in shellfish and could pass to humans eating them **1** It was assumed dilution would make it harmless (illustrates the need for the precautionary principle) – now prohibited and replaced by copper **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	3
05.3	**1 mark each** 1 mark for each of the 3 points below. **1** In the low-oxygen conditions of the spoil heap iron is in its soluble reduced form and is leached into the river **1** In the river it is oxidised to insoluble iron oxide (orange sediment) – the oxidation uses dissolved oxygen so the river is deoxygenated, killing aerobic organisms **1 Control:** drainage water is passed over mesh screens where the iron oxidises and is deposited before the water reaches the river – the solid iron is removed periodically **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leached** → nutrients or ions washed down out of the soil / washed away by water	3
05.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Lead-acid car batteries (recycled) **1** Lead flashing on roofs / windows **1** Radiation shielding **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
05.5	1 mark for the point below. **1** They inhibit enzyme function, especially the enzymes of the nervous system / nerve cells **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	1 mark for the point below. **1** The length of time a pollutant remains in the environment before it is broken down / degraded (chemically or biologically) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time	1
06.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1 Light:** UV light degrades many pesticides (eg neonicotinoids break down fastest in sunlight) so persistence is shorter in bright conditions **1 Temperature:** higher temperatures increase the rate of chemical breakdown / microbial decomposition **1 Soil microbes / oxygen:** aerobic microbial activity breaks pesticides down so persistence is shorter in well-aerated, biologically active soils **1** pH / presence of other chemicals may increase or decrease the rate of degradation **1** Adsorption onto clay / organic matter may protect from degradation and increase persistence **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time • **Adsorption** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-266; AQA spec 3.4.3.2.8.	
07.1	**1 mark each** 1 mark for each of the 2 points below. **1** About 8 nights **1** The running mean fluctuates widely with few samples but stabilises at 21-22 from 8 nights onwards – extra samples do not change it significantly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2

Do not write outside the box

Question	Marking guidance	Marks
07.2	**1 mark each** 1 mark for each of the 3 points below. **1** Moth activity varies with temperature, wind, rain and season (adults only fly for part of the year) so catches vary independently of population size **1** Sampling at different conditions / times would make results not comparable **1** Timing between samples should be chosen to detect the temporal variation of interest (eg weekly through the season) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
07.3	**1 mark each** 1 mark for each of the 2 points below. **1 Accuracy:** how close a recorded result is to the real value **1 Precision:** the smallest interval between possible recorded values (eg a ruler marked in mm is more precise than one in cm) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accuracy** → close to the true value • **precision** → readings close together / fine resolution	2
07.4	**1 mark each** 1 mark for each of the 3 points below. **1** A null hypothesis (eg no difference / no correlation) can be tested statistically and accepted or rejected at a stated confidence level **1** Standardised techniques ensure results collected at different times, places or by different people are comparable **1** Other variables are controlled or monitored so their effect can be assessed – allows the study to be repeated / checked (reliability) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **null hypothesis** → a statement that there is no significant difference or correlation • **correlation** → a relationship in which the two variables change together • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393; AQA spec 3.7.1.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 25 + 5 = 30% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • A bare wrong answer with no working. • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	1
08.2	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 UV:** short wavelength – mostly absorbed in the stratosphere by monatomic, diatomic and triatomic oxygen (ozone layer) which are formed and destroyed in a dynamic equilibrium – converted to chemical energy in these reactions – little reaches the surface **2 Visible light:** transmitted through the atmosphere easily (some reflected by clouds / dust) – absorbed by the Earth's surface and converted to heat – some absorbed by chlorophyll and converted to chemical energy in photosynthesis **2 IR:** emitted by the warm surface (long-wave) – absorbed by greenhouse gases (CO2, water vapour, methane) and converted to heat – the warm atmosphere re-emits IR to the surface and reduces conduction losses – some escapes to space **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV • **long-wave** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	6

Do not write outside the box

Question	Marking guidance	Marks
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Greenhouse gases absorb most of the IR emitted by the surface and re-radiate it in all directions, including back down **1** The surface is repeatedly warmed and re-emits, so the total IR exchanged between surface and atmosphere exceeds the incoming solar energy – only the net loss (70%) balances the absorbed solar energy at equilibrium **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2
08.4	1 mark for the point below. **1** The atmosphere forms layers (troposphere, stratosphere, mesosphere) with different temperature gradients because energy is absorbed at different altitudes (eg UV heating the stratosphere, surface heating the troposphere) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 87-89; AQA spec 3.2.1.1.	
09.1	**1 mark each** 1 mark for each of the 2 points below. **1** Occupancy sensors detect infrared / movement and switch lights or appliances off when no one is in the room **1** Programmable thermostats set different temperatures for different rooms and times so unused rooms / periods are not heated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2
09.2	**Marks shared** Up to 3 marks in total - share them between the 2 points below according to how completely each one is made. **up to 2** LEDs convert a much larger proportion of electricity to light rather than heat **up to 2** Washing machines with cold / low-temperature cycles and fast spins (less drying) – dishwashers using less water so less heating – cookers with double / triple glazed doors – fridges with linear compressors – LED screens using half the power of plasma – germanium transistors **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
09.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Turning off lights / appliances rather than standby **1** Turning thermostats and radiator valves down / off in unused rooms **1** Only heating the water needed (kettle) **1** Showering instead of bathing – eco-cycles **1** Choosing smaller vehicles / less travel **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
09.4	**1 mark each** 1 mark for each of the 2 points below. **1** Stored hot water loses heat continuously through the tank walls and pipes **1** Heating as needed (or just before) avoids these standing losses – if storage is needed the tank should be well insulated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 232-233; AQA spec 3.3.4.2.2.	
10.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** No reservoir range = 3.9 - 0.7 = 3.2 m – with reservoir = 2.7 - 1.7 = 1.0 m **1** Reduction = 2.2 m **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (m) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2

Do not write outside the box

Question	Marking guidance	Marks
10.2	**1 mark each** 1 mark for each of the 2 points below. **1** After heavy rainfall surplus water is stored, reducing peak flows / flooding downstream **1** During dry periods the dam is opened to release water and maintain river levels **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
10.3	**1 mark each** 1 mark for each of the 2 points below. **1** Deforestation / urbanisation reduce infiltration and interception **1** So runoff reaches the river faster after rain and less water is stored to maintain flow in dry periods **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infiltration** → water soaking into the soil surface **Do not credit** • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Site factors:** topography - narrow exit from a large deep basin so a small dam holds a large volume – impermeable, strong bedrock without faults or seismic activity – large catchment with regular high rainfall and low evaporation – catchment land use with low pollution risk (pesticides, industry) and low soil erosion to avoid sedimentation – forest / vegetation in the flooded area decaying anaerobically releasing methane – access for workers and materials and distance to demand – value of the land lost (urban, conservation, agricultural) - loss-benefit analysis differs between countries • **Alternatives for supply:** rainwater collection, rivers, estuary barrages, desalination (energy-intensive so only where freshwater is inadequate), unexploited aquifers (N Africa, Kenya), inter-basin transfers (China, Australia, Wales-England), artificial aquifer recharge using surplus wet-season water, river-regulation reservoirs, afforestation to even out flow • **Conservation strategies:** low-water appliances, dual-flush toilets, xeriscaping, drip irrigation, grey-water re-use, leak repair (about 20% lost in the UK), metering • Treatment / pollution control allowing re-use of contaminated sources: sewage treatment, buffer strips, mine drainage control • **Evaluation:** reservoirs give large reliable supplies and flood control but flood land, cause habitat loss and methane – conservation is cheap and quick but limited by behaviour – transfers create conflicts between regions – desalination has high energy / carbon cost – combination needed as demand grows • Justified conclusion	9

Do not write outside the box

Question	Marking guidance	Marks
	Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **sustainability** → meeting present needs without stopping future generations meeting theirs • **impermeable** → does not let water through • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	
	Model answer - Level 3 (8 marks)	
	A reservoir site is judged first on topography: a narrow exit from a large, deep basin lets a small, cheap dam hold a large volume. The bedrock must be impermeable so water is not lost and strong enough, without faults or seismic activity, to carry the dam. The catchment should be large with regular, high rainfall and low evaporation, and its land uses should not pose a pollution risk from industry or pesticides or cause the soil erosion that would silt the reservoir. Flooding forest or vegetation causes anaerobic decay and methane. Access for workers and materials, distance to the area of demand, and the value of the land lost - urban areas and nature reserves protected, farmland less so - complete a loss-benefit analysis that different countries weigh differently.	
	Supply can also be increased without new reservoirs: rainwater collection in cities, estuary barrages, desalination where freshwater is genuinely inadequate, newly found aquifers in North Africa and Kenya, inter-basin transfers as in China and from Wales to England, artificial recharge of aquifers with wet-season surplus, river-regulation reservoirs and afforestation to even out flow. Demand can be cut by low-water appliances, xeriscaping, drip irrigation, grey-water re-use, metering and repairing the leaks that lose about a fifth of UK supply, and treatment lets contaminated sources be re-used.	
	Reservoirs give large, reliable supplies and flood control but flood land and habitats and release methane; conservation is cheap and quick but limited by behaviour; transfers create conflicts between regions; desalination carries a high energy and carbon cost. As demand grows, the sustainable answer is conservation first, then a combination of sources matched to each region.	
	Weaker answer - Level 2 (4 marks)	
	A reservoir needs a large catchment area with lots of rain, impermeable rock so the water does not drain away, and a narrow valley so the dam is small. It should be near the people who need the water and not flood important land. Other ways to get water include desalination, which is expensive, and collecting rainwater. Water can be conserved by fixing leaks, using water meters and having low-flush toilets. All of these help make water supplies more sustainable.	
	What makes the difference	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 2 lists correct site factors but without the reasons (why impermeable, why the valley shape matters, why land use in the catchment matters, methane from flooded vegetation).	
	• Level 3 covers far more alternatives (aquifer recharge, river regulation, transfers, afforestation) and evaluates each; Level 2 evaluates only desalination.	
	• Level 3 ends with a prioritised judgement; Level 2 ends with 'all of these help'.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-127; AQA spec 3.2.2.4-3.2.2.5.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Properties:** state of matter, density, persistence, toxicity, specificity, reactivity, adsorption, solubility, bioaccumulation / biomagnification, synergism, mutagenic / carcinogenic / teratogenic action, mobility • **How properties predict behaviour:** dispersal (gases vs solids, density - Bhopal), how far and how long (persistence, mobility), where they concentrate (adsorption on sediments, food chains) • Environmental features interacting with properties: temperature inversions, currents, pH, oxygen, light, adsorbent materials • **Control based on properties:** persistent / liposoluble pesticides banned (organochlorines) and replaced by non-persistent / specific ones – systemic pesticides and seed coatings • **Heavy metals:** lead uses discontinued, storage at high pH to reduce solubility – mercury chemical form – radioactive waste categorised by half-life / activity and contained (encapsulation, sealed storage) • Reactive primary pollutants controlled to prevent secondary pollutants (catalytic converters, FGD, urea sprays) – oil properties (floats, insoluble) exploited by booms and skimmers	25

Do not write outside the box

Question	Marking guidance	Marks
	– density in cyclone separators • **Principles:** CPA and CGM based on properties – emission location / timing – ALARA and BATNEEC – polluter pays – precautionary principle • **New technologies:** satellite oil-spill monitoring, GPS, polymer adsorption, phytoremediation, bioremediation • **Evaluation:** prevention better than cure – limits of remediation – cost-effectiveness – contrast success of ozone (Montreal) with climate change – judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-245, 262-266, 275-285, 295-304; AQA spec 3.4.1-3.4.3.	
	Model answer - Level 5 (22 marks)	
	Pollutants cause damage in proportion to how much reaches sensitive organisms and how harmful it is when it gets there. Both depend on properties, so knowing a pollutant's properties lets us predict where it will go and which control will work.	
	State of matter and density decide dispersal. Solid particulates settle near the source unless they are very small: PM10 and smaller stay suspended, travel far and penetrate lungs, so smoke is controlled at the point of release with electrostatic precipitators, cyclones and bag filters, and smokeless zones prevent its production. Gases disperse widely, and dense gases such as the hydrogen cyanide released at Bhopal hug the ground, so location and emission height matter. Persistence decides whether dilution is acceptable. Sewage and other biodegradable wastes break down, so treatment plus dilution in a river or the sea is sufficient; organochlorine pesticides, CFCs, heavy metals and long-lived radioisotopes are not broken down, so they travel, accumulate and must be contained, banned or replaced - DDT and CFCs were both controlled by substitution once their persistence was understood.	
	Solubility governs the pathway. Water-soluble nitrate leaches into rivers and aquifers, so it is controlled by nitrate vulnerable zones, fertiliser timing and buffer strips; liposoluble pollutants such as DDT and methyl mercury pass into fat and are biomagnified along food chains, as at Minamata, so even low concentrations in water are dangerous and the only effective control is to stop release. Adsorption on clay immobilises caesium and metals, which is why radioactive discharges are monitored in sediments and why activated carbon and polymer adsorbents are used deliberately in effluent treatment.	

Do not write outside the box

Question	Marking guidance	Marks
	Toxicity, specificity and mode of action set acceptable doses and identify species at risk. Neonicotinoids are toxic to insects but not vertebrates because they cannot cross the vertebrate blood-brain barrier, which explains both their popularity and their effect on bees; pyrethroids replaced organochlorines because they are less persistent. For mutagens and carcinogens such as ionising radiation there is no safe dose, so exposure is kept As Low As Reasonably Achievable with shielding, distance, time limits and monitoring. Reactivity explains secondary pollutants - sulfur dioxide oxidised to sulfuric acid, nitrogen oxides and hydrocarbons forming ozone and PANs in photochemical smog - so control targets the primary emissions with flue-gas desulfurisation and catalytic converters. Synergism between smoke and sulfur dioxide, or ozone and sulfur dioxide, means limits must consider pollutants together.	
	Properties interact with the environment: temperature inversions trap pollutants, currents disperse them, pH and oxygen alter solubility. Critical Pathway Analysis combines the two to predict where a discharge will concentrate and to focus monitoring, and Critical Group Monitoring checks the most exposed people. These principles have reduced smoke, sulfur dioxide, lead and organochlorines dramatically in developed countries. Their limits are pollutants with diffuse sources or that underpin economies - nitrate, carbon dioxide - where knowing the properties is not the problem; changing behaviour is. Even there, the precautionary principle applies the same logic: predict the properties of a new substance from similar ones and control it before release, because prevention is always cheaper than remediation.	
	What makes the difference	
	• Every paragraph is organised around a **property**, each linked to a prediction and a named control - the essay answers the question throughout rather than describing pollutants.	
	• Examples span the specification (smoke, acid rain, pesticides, heavy metals, radiation, nutrients, oil), with quantitative and named detail (Bhopal, Minamata, PM10, ALARA), which is what Level 5 breadth means.	
	• The final paragraph evaluates: where the approach has worked, where it has not and why, and reaches a justified conclusion.	
	• A Level 3 version of this essay would describe several pollutants and their effects accurately but only occasionally connect a property to a control, and would end without a judgement.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Features of fertile soil:** water, nutrients, air, organic matter, pH, biota, texture, structure, depth – slow rate of soil formation vs rapid erosion (dynamic equilibrium broken) • **Causes of degradation:** vegetation removal, ploughing vulnerable soils, overgrazing, reduced organic matter and biota, cultivating steep slopes, compaction, salinisation from irrigation, nutrient depletion, pollution • **Types of erosion:** rain splash, runoff, wind, slumping – consequences: reduced productivity, river and reservoir sedimentation, flooding, coastal sedimentation, particulates, desertification, landslides • USLE as a tool for estimating loss and testing management • **Management:** long-term crops, low / zero tillage, contour ploughing, tied ridging, terracing, stone rows, windbreaks, multicropping / strip cropping, mulching, organic matter, livestock control, drainage / pH / nutrient management, conservation of soil biota • Links to sustainable agriculture, carbon storage, hydrological cycle, circular economy	25
	• Evaluation and judgement on which strategies are most effective in different environments *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169, 328-337; AQA spec 3.2.5, 3.5.1.3, 3.5.1.5.	

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